Haoran Guo

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Education

Tsinghua University August 2024 – present

- Ph.D. Student in Management Science and Engineering
- Advisor: Wei Qi

Cornell University August 2026 – present

- Visiting Graduate Researcher, SC Johnson College of Business
- Faculty Host: Meng Qi

Tsinghua University August 2020 – June 2024

- B.E. in Industrial Engineering (GPA: 3.94/4.00, Rank: 3/67)

Research

Research Interests

- Broad areas: sustainable urban operations, data-driven optimization, analytics for social good
- Primary focus: walkable city operations

Working Papers

1. **Haoran Guo**, Sentao Miao, Wei Qi, Jian Cui, Zhenning Dong. Designing Walkable Neighborhoods: A Data-Driven Optimization Framework. Major Revision at ***Management Science***. [Link]
   - **Winner**, CSAMSE Best Paper Award, 2026.
   - **Finalist**, INFORMS Public Sector Operations Research (PSOR) Best Paper Award, 2025.
   - **Media Coverage**: CU Boulder Today; Amap (in Chinese); Mr. Mall (in Chinese).
2. **Haoran Guo**, Meng Qi, Wei Qi. Overworking or Outsourcing: Integrated Prediction and Optimization for Managing Overtime under Uncertainty. Reject and Resubmit at ***Management Science***. [Link]
   - **Winner**, INFORMS Transportation Science & Logistics (TSL) Best Student Paper Award, 2025.
   - **Finalist**, IISE DAIS Division Best Student Paper Competition, 2024.

Publications

1. Yiwei Wu, **Haoran Guo**, Jingwen Qi, Shuaian Wang, and Lu Zhen. Ship Refueling Optimization for Dual-Fuel Ships Considering Carbon Intensity Indicator Rating Limit and Uncertain Fuel Prices. *Multimodal Transportation* 3.3 (2024): 100138. [Link]
2. Wei Wang, **Haoran Guo**, Fei Li, Lu Zhen, and Shuaian Wang. A Bi-Level Programming Approach to Optimize Ship Fouling Cleaning. *Journal of Marine Science and Engineering* 11.12 (2023): 2324. [Link]

Honors & Awards

Research Awards

- Winner, CSAMSE Best Paper Award 2026
- Second Place, INFORMS Data Mining Society (DMS) Data Challenge (team led by Lingjie Zhao) 2026
- Winner, INFORMS Transportation Science & Logistics (TSL) Best Student Paper Award 2025
- Finalist, INFORMS Public Sector Operations Research (PSOR) Best Paper Award 2025
- Finalist, IISE DAIS Division Best Student Paper Competition 2024

Grants & Scholarships

- INFORMS Student Scholarship ($1,500) 2026
- INFORMS TSL Doctoral Grant ($3,000) 2025

Undergraduate Honors

- Outstanding Undergraduate, Tsinghua University and Beijing Municipality 2024
- China National Scholarship 2022
- Tsinghua University Scholarship for Comprehensive Excellence 2021
- Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2021
- Gold Award, Tsinghua IE Summer Field Research Program 2021

Conference Presentations

- ASCI International Conference, Guangzhou, China (forthcoming) January 2027
- INFORMS Annual Meeting, San Francisco, USA November 2026
- INFORMS Optimization Society Conference, Atlanta, USA March 2026
- POMS-HK International Conference, Shenzhen, China January 2026
- INFORMS Annual Meeting, Atlanta, USA October 2025
- IISE Annual Conference & Expo (online presentation) May 2024

Industry Experience

Unilumin Group Co., Ltd. July 2026 – August 2026

- Research Intern, AI for Education

Service

Academic

- Ad Hoc Referee: *IISE Transactions*, *Production and Operations Management*
- Section Editor, ORAI China (WeChat platform) 2024 – present
- Session Chair, INFORMS Annual Meeting 2026
- Session Co-Chair, POMS-HK 2025

Teaching

- Teaching Assistant, Production Management, Tsinghua iMEM 2026
- Teaching Assistant, Systems Engineering, Tsinghua MEM 2025 – 2026
- AI Teaching Assistant, Strategic Management, Tsinghua MEM 2024